Data Visual & Preprocessing

Parameter	Details
Date	08 March 2026
Team ID	
Project Name	VoltEdge: EV Analytics Portal
Maximum Marks	2 Marks

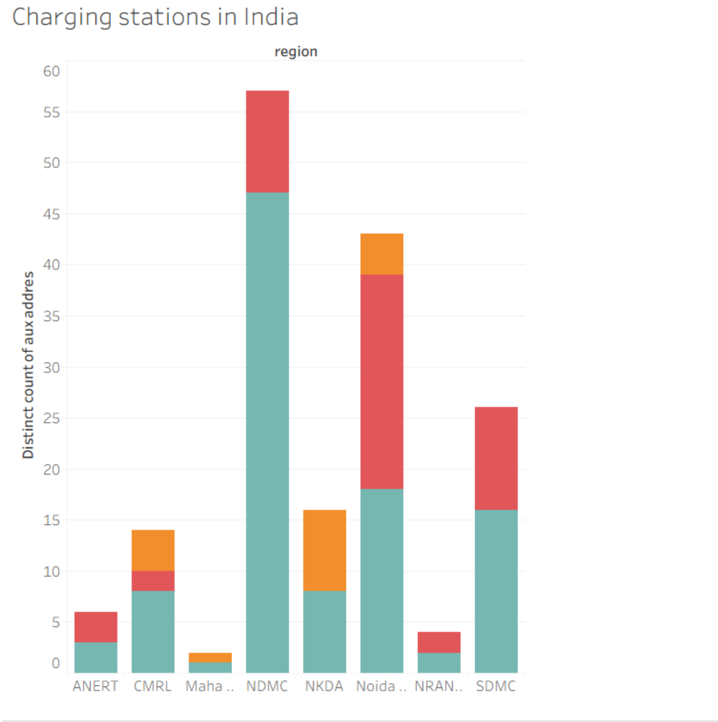
Data Visual & Preprocessing

1. Data Connectivity & Preparation

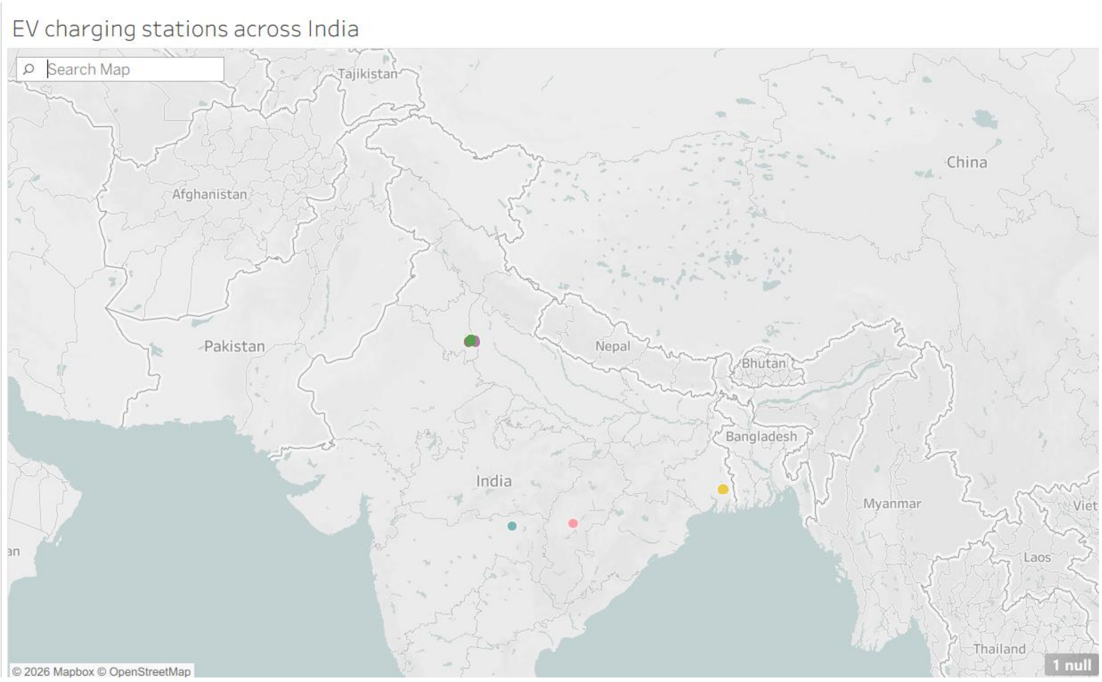
- **Database Integration:** The dataset was stored in a **PostgreSQL** database. A connection was established in Tableau using the PostgreSQL connector, ensuring a structured relational data flow.
- **Preprocessing Steps:**
    - **Data Cleaning:** Used SQL queries in pgAdmin to verify row counts and column integrity. In Tableau, null values in the Price and Range columns were filtered out.
    - **Field Renaming:** Technical headers were renamed (e.g., top_speed_kmh to Top Speed (km/h)) for dashboard clarity.
    - **Data Typing:** Assigned "Geographic Roles" to State/Region fields to enable the **EV Charging Stations Map of India**.
- **Calculated Fields:**
    - **Count:** COUNT([Car]) — to determine the number of models per brand.
    - **Count_powertrain:** COUNT([PowerTrain]) — to analyze the distribution of drive types (AWD, RWD, FWD).

2. Business Questions with Visualizations

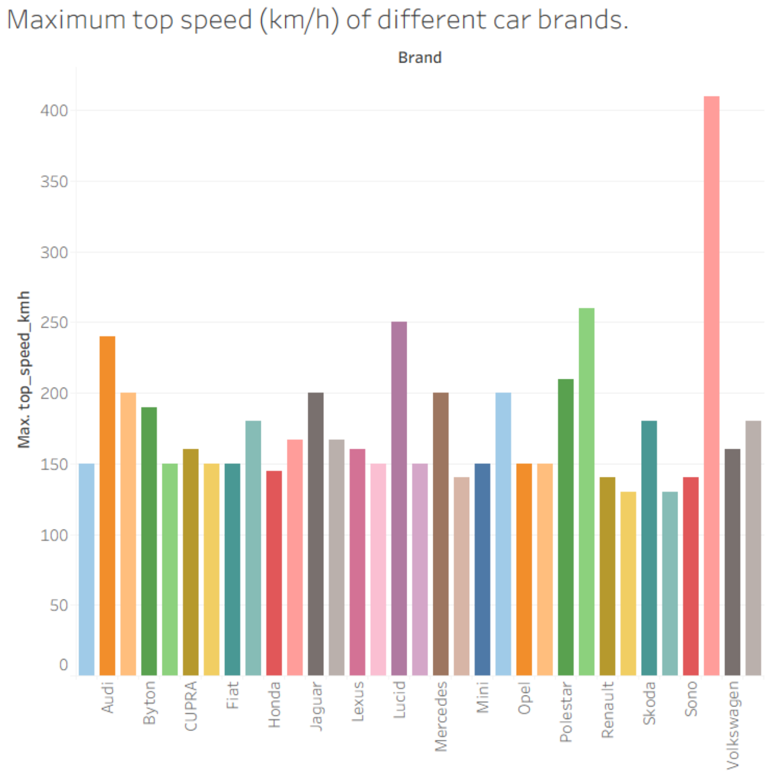
- **Q1: What is the regional distribution of charging types?**



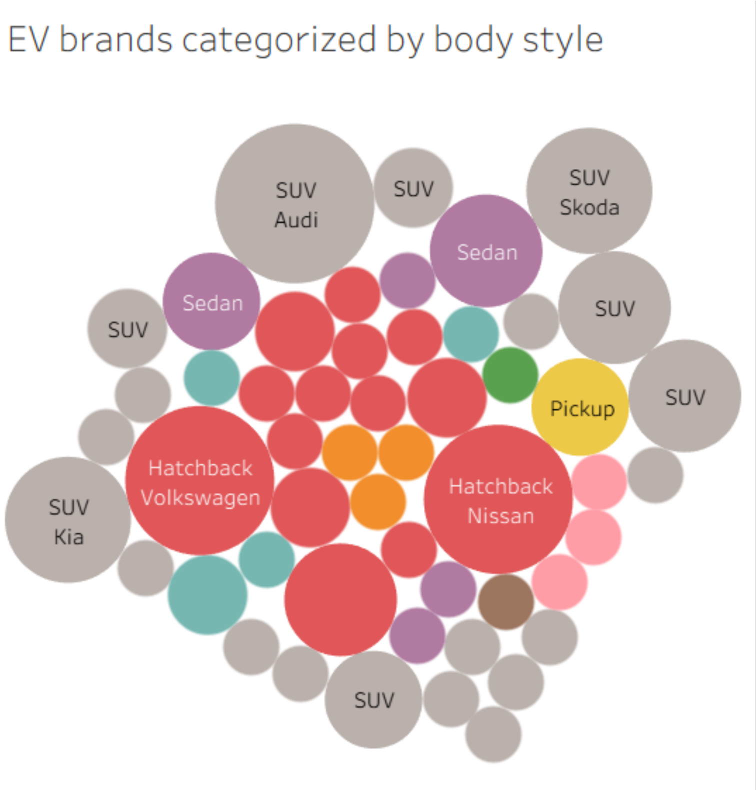
- **Q2: Where are the infrastructure clusters?**



- **Q3: Which brands lead in top-tier performance?**



- **Q4: How does the market split by vehicle category?**



- **Q5: Which brands are the most energy-efficient for fleet use?**